Andreas Hovaldt **Højrup**

MSc in Computer Engineering, specialized in AI, vision and sound
BSc in Robotics

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I recently graduated as a computer engineer, specialized in AI, vision, and sound. I finished my MSc in Computer Engineering in June 2026, with a previous bachelor's in Robotics. My work and university experience spans across a broad range of the ML stack, from applying 2D and 3D visual foundation models to in-field LiDAR perception in my AGCO master's thesis, to approximating model predictive controllers during a five-month internship at CERN, down to the GPU-cluster training, experiment tracking, and the Linux infrastructure underneath. My native language is Danish, but throughout my work and studies, I have also become fluent in English. I am passionate about modern technology and building things for real world problems. I am now seeking ML engineering, computer vision, or MLOps roles where I can solve real problems, contribute to exciting projects, and keep learning.

Work Experience

Jun 2026 – Feb 2026 | **Student Worker | Applied Vision Research, AGCO A/S, Randers, Denmark (hybrid)**

- Trained lightweight task adapters on frozen DINOv3 embeddings to predict machine and material state from in-machine cameras
- Designed a per-instance gating loss that lifted task accuracy from $\sim 70\%$ to over 99%
- Implemented sequence models which beat single-frame (R^2 $0.31 \rightarrow 0.63$)
- Built gradient-rollout explainability tooling

PyTorch | DINOv3 | Vision Foundation Models | Weights & Biases | AI Explainability

Jan 2026 – Sep 2025 | **Computer Engineering Intern, CERN, Geneva, Switzerland**

- Built neural-network approximation of an MPC for HVAC air-handling units, cutting online compute by $\sim 460\times$
- Developed the full pipeline : Apache Spark data extraction, dataset curation, modular PyTorch training, and closed-loop simulation deployment
- Compared MLP, TCN, and transformer models, best reached $R^2 = 0.995$ (regression) and 94% class-averaged accuracy (classification)
- Held zone temperatures within operational bounds in closed-loop run within the EcosimPro digital twin simulation

PyTorch | ONNX Runtime | Transformer | Time series | Apache Spark (NXCALS) | Weights & Biases | EcosimPro

Aug 2025 – Jul 2025 | **Student Worker | 3D Perception Research, AGCO A/S, Randers, Denmark**

- Researched and evaluated point-cloud segmentation techniques for agricultural environments, laying groundwork for the thesis collaboration

Point Cloud Segmentation | Open3D | Pointcept

Nov 2023 – Feb 2019 | **Postal Worker & Assistant Coordinator, PostNord, North Jutland, Denmark**

- Full-time during a sabbatical year, then part-time alongside studies
- Assistant coordinator for weekend parcel delivery across North Jutland municipalities

Education

2024-2026 **MSc in Computer Engineering** with a specialization in AI, vision, and sound at Aalborg University
Overall average : 11.3 Thesis : 12 (Danish 7-point scale)

2021-2024 **BSc in Robotics** at Aalborg University
Overall average : 10.0 Thesis : 10 (Danish 7-point scale)

Publication

Synthetic LiDAR Data Generation using 3D Gaussian Splatting for 3D Semantic Segmentation Jun 2025

Christiansen, A.T.[*], **Højrup, A.H.**[*], Stephansen, M.K.[*], *et al.* ([*]Equal contribution)
Synthetic Data for Computer Vision Workshop, CVPR
arXiv:2506.05009 | OpenReview

Projects

Foundation Models in Agricultural 3D Perception (MSc Thesis, with AGCO) Feb 2026 - Jun 2026

Evaluated 2D and 3D visual foundation models for in-field LiDAR detection and segmentation, on a self-captured, manually annotated agricultural LiDAR dataset alongside ScanNet v2. Domain fine-tuning a 3D point-cloud foundation model raised detection mAP25 by $\sim 4\times$, while injecting frozen DINOv3 image embeddings lifted trained from-scratch models by $\sim 8.3\times$.

PyTorch | 3DETR | DINOv3 | LiDAR | Pointcept | Slurm | Transformer | Detection | Segmentation

Neural Network Approximation of an MPC for HVAC (CERN) Sep 2025 - Jan 2026

Built neural-network approximations of an economic Model Predictive Controller for HVAC air-handling units at CERN's SR1 facility, with a full pipeline from NXCALS/Spark data extraction to closed-loop deployment against an EcosimPro digital twin. The best (transformer) model reached $R^2 = 0.995$ and 94% class-averaged accuracy while cutting online compute by $\sim 460\times$.

PyTorch | Transformers | TCN | Apache Spark | EcosimPro

Personal Sound Zones for Smartphones Feb 2025 - Jun 2025

Deep-learning sound-zone control that forms a private "bright zone" for the phone user and a "dark zone" around them, using only native smartphone speakers and microphones and trained entirely on synthetic room impulse responses. Reached 32.5 dB average acoustic contrast, approaching analytically optimal filters.

PyTorch | Acoustics | Signal Processing | Deep Learning | Pyroomacoustics

Bimanual Manipulation with an Integrated LLM Sep 2023 - Jan 2024

Natural-language-controlled bimanual system using two Kuka IIWA 7 arms, with YOLOv8 object detection and a GPT-4 agent translating prompts into robot commands. Achieved a 73% task success rate while demonstrating LLM-driven adaptive task execution.

LLM Agents | YOLOv8 | GPT-4 | ROS2 | Robotics | LangChain

Personal Projects

Self-Hosted Homelab

Self-hosted home server running 10+ containerized services (Docker/LXC) on Proxmox VE, including reverse-proxied web apps, automated media management, game servers, and monitoring. Secured with a WireGuard VPN and Cloudflare mutual-TLS, with internal services kept behind authenticated access.

Proxmox VE | Docker | LXC | WireGuard | Cloudflare | Linux

Personal Finance Tracker

Full-stack web app that fetches and categorizes bank transactions through the GoCardless API, with a FastAPI + PostgreSQL backend and a React frontend, containerized with Docker Compose and self-hosted.

FastAPI | PostgreSQL | React | GoCardless API | Docker Compose

Technical Skills

Programming	Python (advanced) ; MATLAB, C++, C# (basic)
ML & Vision	PyTorch, scikit-learn, ONNX Runtime, OpenCV, Open3D, Pointcept, Weights & Biases, visual foundation models (DINOv3, Utonia), 3D Gaussian Splatting
Data & Compute	NumPy, SciPy, Pandas, Apache Spark, Slurm GPU clusters (HPC)
Infrastructure	Linux, Docker, Proxmox VE, Git, FastAPI, NGINX, WireGuard
Robotics	ROS2, NVIDIA Isaac Sim, sensor integration (LiDAR, RGB-D cameras)

Interests

Outside of work, I still like to sometimes have something hands-on to tinker with. Previously I have worked on small home electronics projects with either an Arduino or a Raspberry Pi. Recently, I have focused on tinkering with my self-hosted Proxmox setup. When I'm not working on something, I like to go for my weekly runs or, in the weekends, go hiking, and a pleasant evening with friends is always a nice way to unwind.

References

Available upon request.